# Pre-class progress check

## By now, you should have

1. Reviewed slides from weeks 1 — 4
2. Reviewed code from classes 1 — 4
3. Checked assignment 1 (due 10.12)
4. Attempted quizzes 1 - 4 (Quiz 4 released on 09.22)

1. Checked out slides for today
2. Reviewed logistic regression basic slides (released on 09.22).

University of Pittsburgh | School of Public Health

# Supervised learning (II): Classification (I)
# - generative models
# - k-nearest neighbors

## BIOST 2155

Class 5
September 26th, 2025

# The confusion matrix

## Quantifying performance for classification problems

| | |
|---|---|
| **Overall Accuracy:** proportion of total predictions that were correct.<br>*"Overall, how often is the model right?"* | **(TP + TN)/(TP + TN + FP + FN)** |
| **Sensitivity:** proportion of actual positives that were correctly identified.<br>*"Of all the true positive cases, how many did we catch?"* | **TP / (TP + FN)** |
| **Specificity:** proportion of actual negatives that were correctly identified.<br>*"Of all the actual negative cases, how many did we correctly rule out?"* | **TN / (TN + FP)** |
| **Youden's J Index:** summarizes the diagnostic ability, balancing sensitivity and specificity | **Sensitivity + Specificity - 1**<br>*(Express using TP, TN, FP, FN)* |

| Classify as $\hat{Y} = 1$ if $\hat{P}(Y = 1 \mid X = x) > 0.5$ | | *Predicted Label* | |
|---|---|---|---|
| | | *Positive* | *Negative* |
| **Actual label** | **Positive** | **T**rue **P**ositive | **F**alse **N**egative |
| | **Negative** | **F**alse **P**ositive | **T**rue **N**egative |

# The confusion matrix (contd.)

## Quantifying performance for classification problems

What if we lower the threshold to 0.4?

- Sensitivity would increase.
- Specificity would likely decrease.

How do we evaluate a model's performance across all possible thresholds?

**Receiver Operating Characteristic (ROC) Curve**

| Classify as $\hat{Y} = 1$ if $\hat{P}(Y = 1 \mid X = x) > 0.5$ | | *Predicted Label* | |
|---|---|---|---|
| Classify as $\hat{Y} = 1$ if $\hat{P}(Y = 1 \mid X = x) > 0.4$ | | *Positive* | *Negative* |
| **Actual label** | **Positive** | True Positive | False Negative |
| | **Negative** | False Positive | True Negative |

fig1

# The confusion matrix (contd.)

## Quantifying performance for classification problems

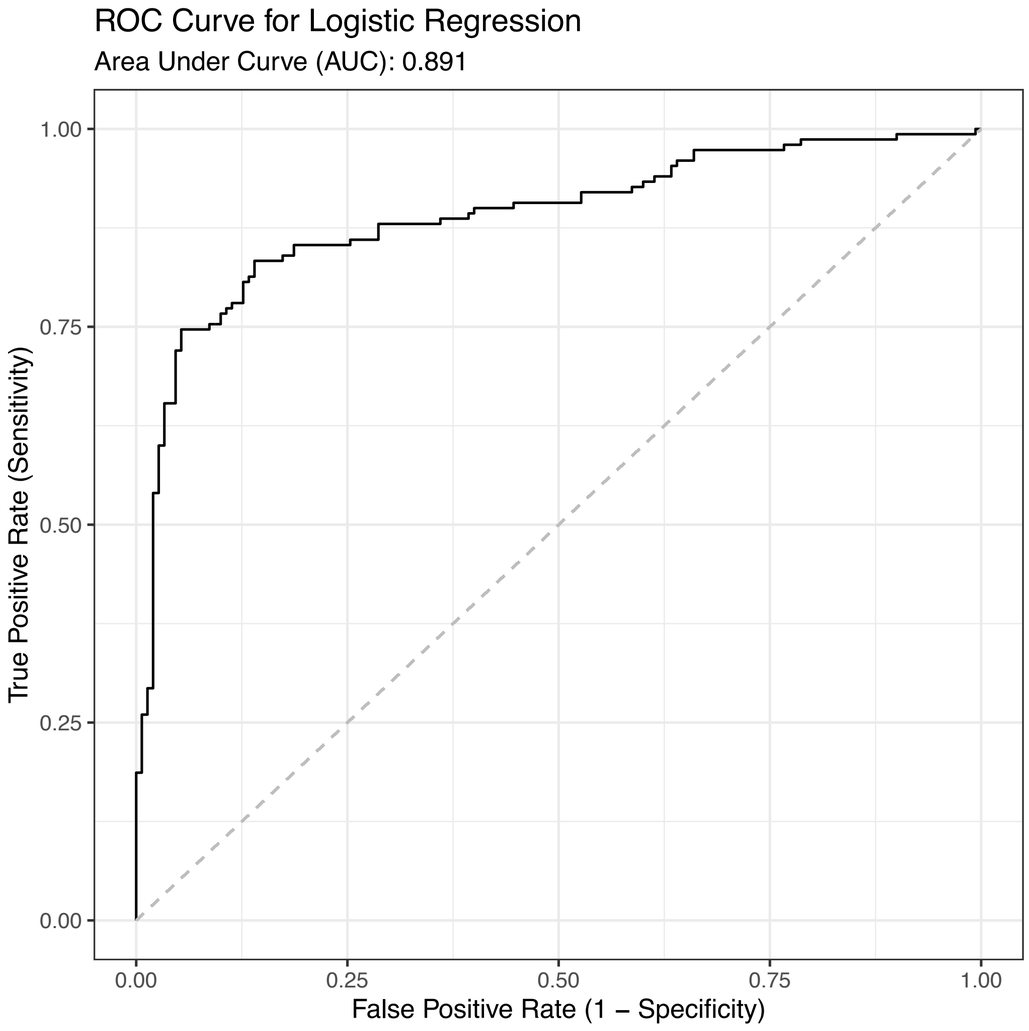
ROC curve visualizes the performance of a classifier by plotting the **True Positive Rate** (Sensitivity) against **False Positive Rate** (1 - Specificity) at every possible threshold.

Y-axis: **Sensitivity** vs X-axis: **1 - Specificity**

**A better model has a curve that is closer to the top-left corner (high sensitivity, low false positive rate).**

**AUC: Area Under the (ROC) Curve.** *Measures the entire two-dimensional area underneath the entire ROC curve. It provides an aggregate measure of performance across all classification thresholds.*

- **AUC = 1:** A perfect classifier.
- **AUC = 0.5:** A model with no discrimination ability, equivalent to random guessing (the diagonal line).
- **0.5 < AUC < 1:** The model has some ability to discriminate. The closer the AUC is to 1, the better the classifier.



# Generative models for classification

## Why not just stick with logistic (multinomial) regression?

- In logistic (or multinomial) regression approach, we model

  $P(Y = k \mid \boldsymbol{X})$ for $k \in \mathscr{Y}$, $\mathscr{Y}$ is the set of values that $Y$ can take.

- In the generative model approach, we
  - **Step 1:** Model $P(\boldsymbol{X} \mid Y = k)$ separately for each value of $k \in \mathscr{Y}$
  - **Step 2:** Use <u>Bayes theorem</u> to 'flip' step 1 into estimates for $P(Y \mid \boldsymbol{X})$

- For substantial separation between the classes, $\hat{\boldsymbol{\beta}}_{LOGIS}$ are unstable.
- If $\boldsymbol{X}$ is <u>approximately</u> normal for each $k \in \mathscr{Y}$, then this approach is better for small $n$

# Generative models for classification

## Recap of Bayes theorem

- In the generative model approach, we
  - **Step 1:** Model $P(\boldsymbol{X} \mid Y = k)$ separately for each value of $k \in \mathcal{Y}$
  - **Step 2:** Use <u>Bayes' theorem</u> to 'flip' step 1 into estimates for $P(Y = k \mid \boldsymbol{X})$

- Bayes' theorem

$$\mathbb{P}(Y = k \,|\, \boldsymbol{X} = \boldsymbol{x}) = \frac{\mathbb{P}(\boldsymbol{X} = \boldsymbol{x} \,|\, Y = k) \cdot \mathbb{P}(Y = k)}{\mathbb{P}(\boldsymbol{X} = \boldsymbol{x})} = \frac{\mathbb{P}(\boldsymbol{X} = \boldsymbol{X} \,|\, Y = k) \cdot \mathbb{P}(Y = k)}{\sum_{k \in \mathcal{Y}} \mathbb{P}(\boldsymbol{X} = \boldsymbol{X} \,|\, Y = k) \cdot \mathbb{P}(Y = k)}$$

  - $\mathbb{P}(\boldsymbol{X} = \boldsymbol{X} \,|\, Y = k) = f_k(\boldsymbol{x})$ describes density of predictors in class (label) $k$
  - $\mathbb{P}(Y = k) = \pi_k$ is the marginal probability of class (label) $k$

fig2

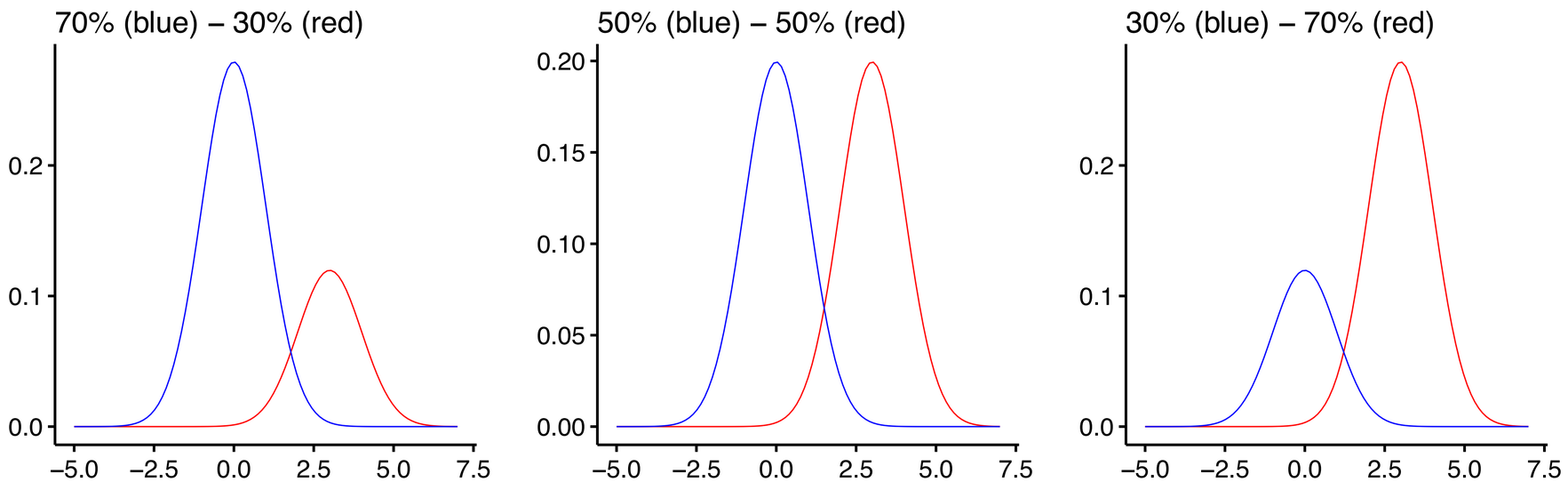
# Generative models for classification

## Recap of Bayes theorem (contd.)

- **Step 1**: Model $P(\boldsymbol{X} \mid Y = k)$ separately for each value of $k \in \mathscr{Y}$

- **Step 2**: Use Bayes' theorem to 'flip' step 1 into estimates for $P(Y = k \mid \boldsymbol{X})$

- **Step 3**: Predict label $\hat{k}(\boldsymbol{x}) = \text{argmax}_k \, \mathbb{P}(Y = k \mid \boldsymbol{X} = \boldsymbol{x})$




Classify to the highest density!!

# Linear discriminant analysis

**Starting step: $p = 1$ and using Gaussian distributions for $f_k(x)$**

- Starting step: $f_k(x) = \sigma^{-1}\phi((x - \mu_k)/\sigma)$ (**can be other distributions as well**)
  - $X \mid Y = k \sim$ normal; **different means (<u>known</u>), same variance (<u>known</u>)**.
    - So, $p_k(x) \propto \pi_k \exp(-(x - \mu_k)^2/2\sigma^2)$
  - **Decision rule:** Assign to class $k_0$ if $p_{k_0}(x) > p_k(x)$ for all $k \neq k_0$
    - $p_{k_0}(x) > p_k(x) \iff \delta_{k_0}(x) > \delta_k(x)$
    - **Discriminant function** $\delta_k(x) = x\dfrac{\mu_k}{\sigma^2} - \dfrac{\mu_k^2}{2\sigma^2} + \log(\pi_k)$

For real data: we use plug-in estimates

$$\hat{\pi}_k = n_k/n \qquad \hat{\mu}_k = \frac{1}{n_k}\sum_{i:y_i=k} x_i \qquad \hat{\sigma}^2 = \sum_{k=1}^{K} \frac{n_k - 1}{n - K}\hat{\sigma}_k^2$$

# Linear discriminant analysis

**Next: General $p$ and using Gaussian distributions for $f_k(\boldsymbol{x})$**

$$f_k(\boldsymbol{x}) = \frac{1}{(2\pi)^{p/2} \, |\boldsymbol{\Sigma}|^{1/2}} e^{-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu}_k)^T \boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu}_k)}$$

$X \mid Y = k \sim$ **multivariate** normal. **Different means** and **same variance.**

**But for a set of covariates, not just one**

$$\delta_k(\boldsymbol{x}) = \boldsymbol{x}^T \Sigma^{-1} \mu_k - \frac{1}{2} \mu_k^T \Sigma^{-1} \mu_k + \log \pi_k$$

For real data, we use plug-in estimates (again)

fig3

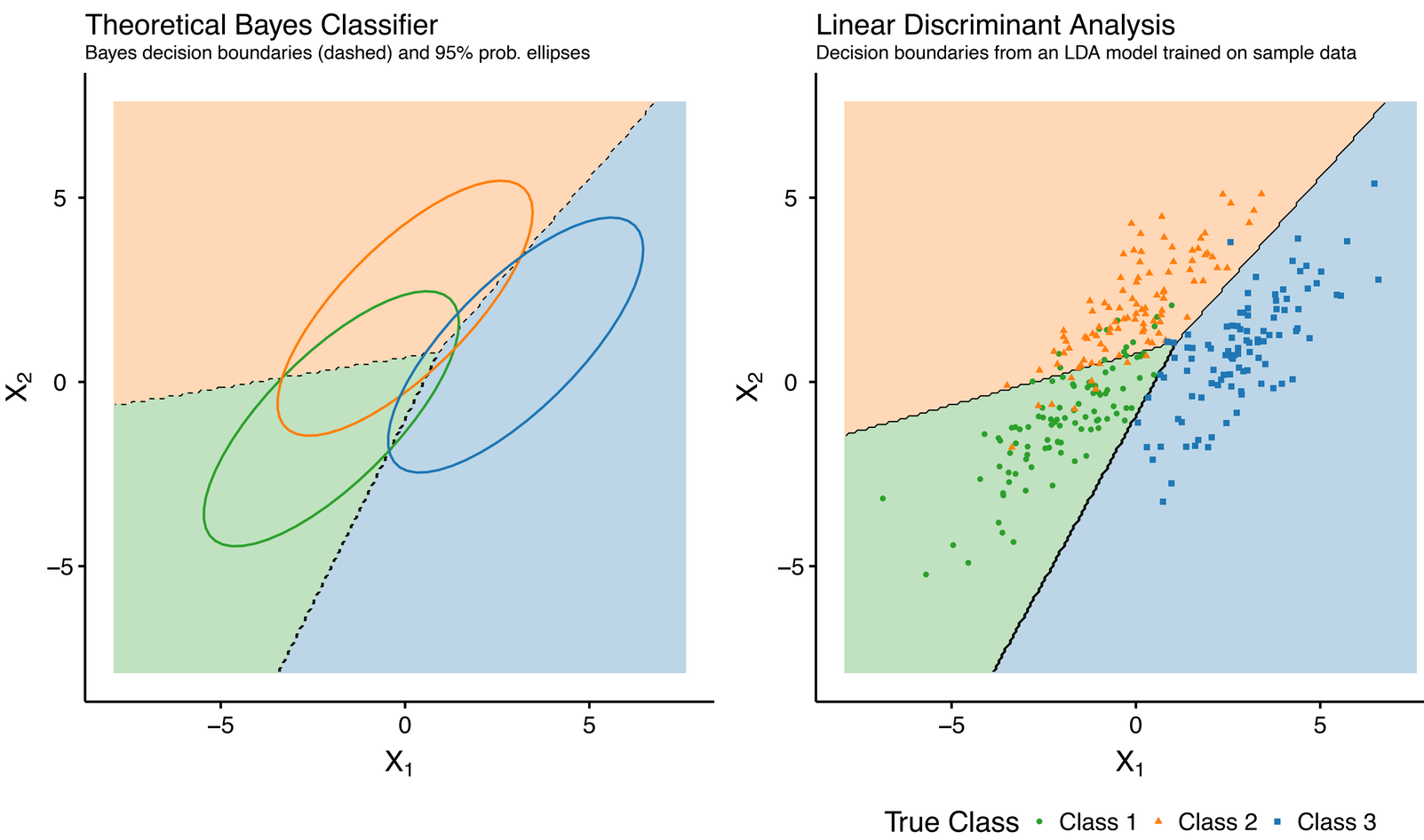
# Linear discriminant analysis

## Illustration: General $p$ and using Gaussian distributions for $f_k(\boldsymbol{x})$

Three distinct classes (green, orange, blue), each with a bivariate Gaussian distribution. Each class has its own mean (center) but share a common covariance matrix

Left panel: theoretical "best case" classifier, known as the Bayes decision boundary (dashed lines). **This ideal boundary is calculated using the true, known mean and covariance of each class.**

Right panel: the true parameters are unknown. LDA model is trained on data. The resulting linear lines are the decision boundaries estimated by the LDA model.



# Quadratic discriminant analysis

**General $p$, using Gaussian distributions for $f_k(\boldsymbol{x})$**

$$f_k(\boldsymbol{x}) = \frac{1}{(2\pi)^{p/2}\,|\,\boldsymbol{\Sigma}_k\,|^{1/2}} e^{-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu}_k)^T\boldsymbol{\Sigma}_k^{-1}(\boldsymbol{x}-\boldsymbol{\mu}_k)}$$

$X \mid Y = k \sim$ **multivariate** normal. **Different means** and **different variances.**

$$\delta_k(\boldsymbol{x}) = -\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu}_k)^T\Sigma_k^{-1}(\boldsymbol{x}-\boldsymbol{\mu}_k) - \frac{1}{2}\log|\,\Sigma_k\,| + \log\pi_k$$

For real data, we use plug-in estimates (again)

fig4

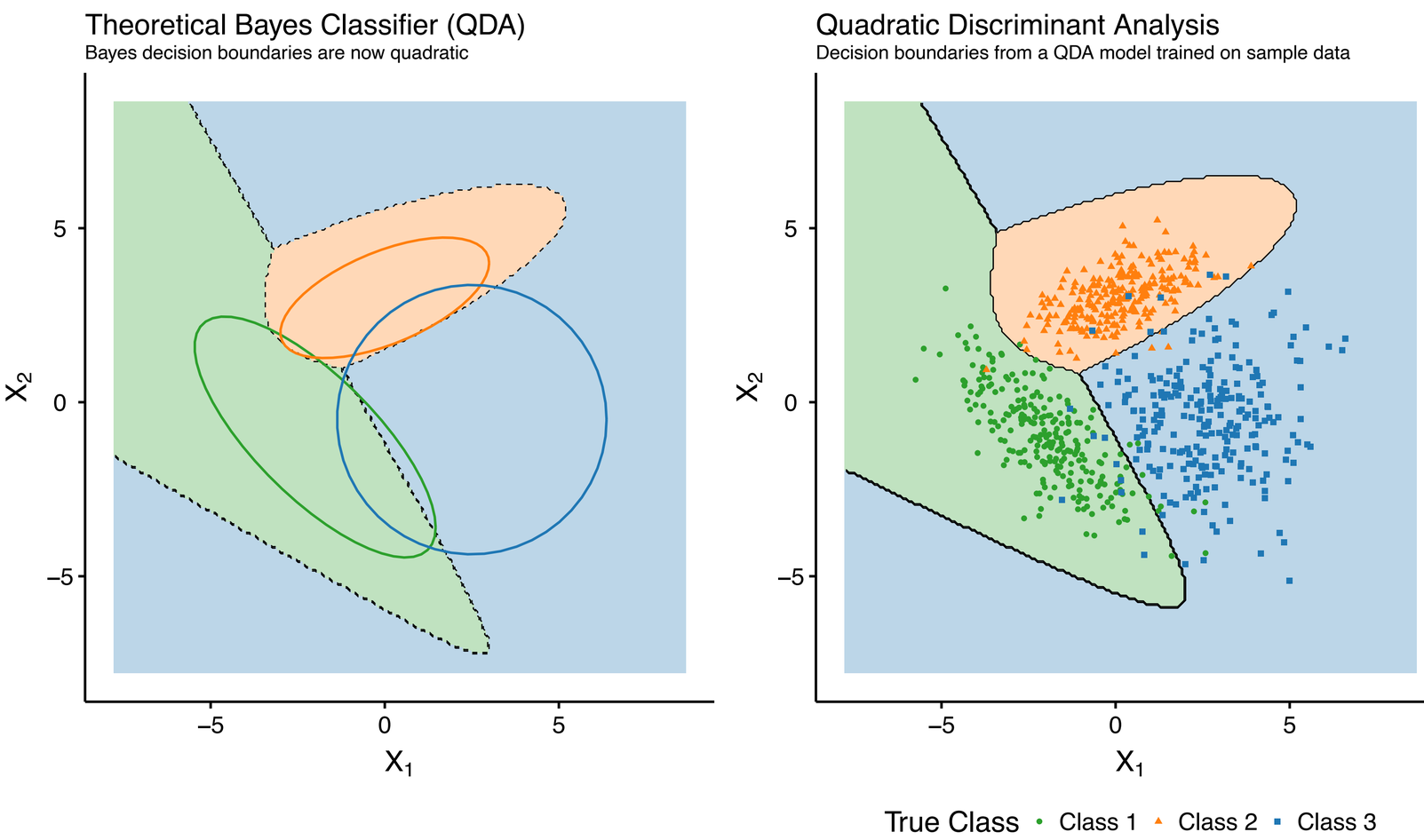
# Quadratic discriminant analysis

## Illustration: General $p$, using Gaussian distributions for $f_k(\boldsymbol{x})$

Three distinct classes (green, orange, blue), each with a bivariate Gaussian distribution. Each class has its own unique mean (center) and covariance matrix

Left panel: theoretical "best case" classifier, known as the Bayes decision boundary (dashed lines). **This ideal boundary is calculated using the true, known mean and covariance of each class.**

Right panel: the true parameters are unknown. QDA model is trained on data. The resulting parabolic curves are the decision boundaries estimated by the QDA model.



# k-Nearest Neighbors

## A non-parametric classification *(and regression)* tool

- For a new observation $(\boldsymbol{x}_0, y)$, it finds the $K$-most similar training observations (i.e., neighbors) and makes a prediction based on them.
- **How it works:**
  - **Step 1:** fix $K$.
  - **Step 2:** find neighbor set $\mathcal{N}_0$. *Identify the $K$ training data points that are <u>closest</u> to the new, unseen observation $\boldsymbol{x}_0$*
  - **Step 3:** majority vote. Assign the new observation to the class that is most common among its $K$ neighbors.

**Non-parametric:** It makes no assumptions about the functional form of the boundary. Can find highly non-linear boundaries.

Instance-based: No model is explicitly trained; it simply uses the entire training dataset at prediction time.

$$\mathbb{P}(Y = j \mid X = x_0) = \frac{1}{K} \sum_{i \in \mathcal{N}_0} I(y_i = j)$$

$$\hat{Y} = \text{argmax}_{j \in \mathcal{Y}} \, \mathbb{P}(Y = j \mid X = x_0)$$

fig5

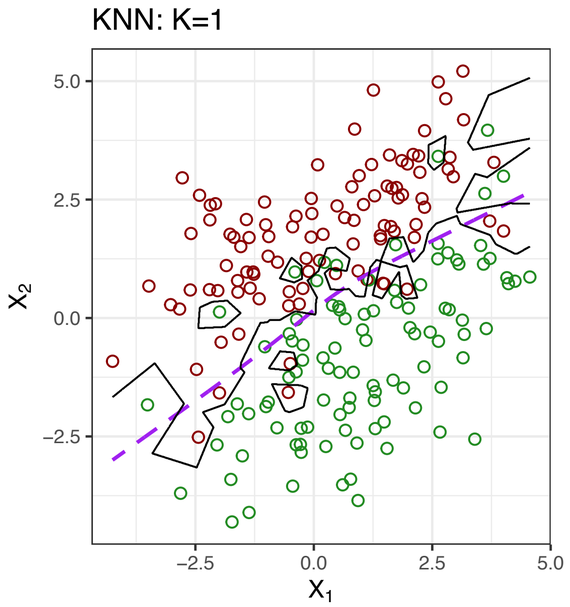
# k-Nearest Neighbors (contd.)

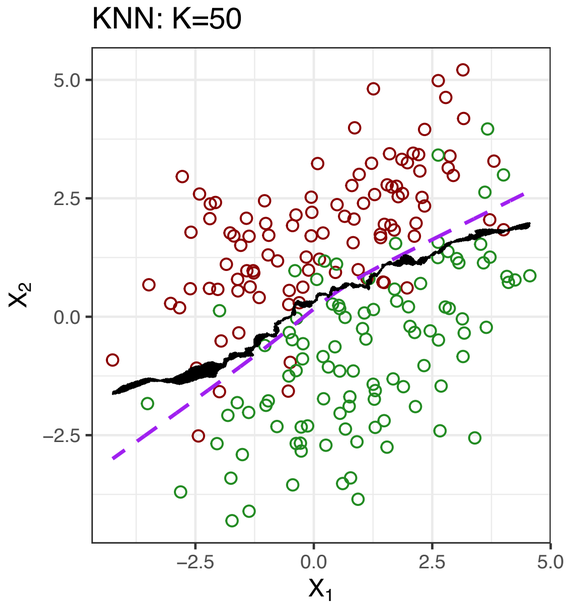
## Choosing K and the Bias-Variance Trade-Off

- Low K (e.g., K=1):
  - **High Flexibility:** The decision boundary is very jagged and follows the training data closely.
  - **Low Bias, High Variance:** The model is prone to overfitting by capturing noise in the data.

- High K (e.g., K=100):
  - **Low Flexibility:** The decision boundary becomes much smoother and resembles a linear boundary.
  - **High Bias, Low Variance:** The model might be too simple and underfit the data.

- How to Choose the Optimal K: **cross-validation!!**



# Post-class debrief

## Today we discussed…

This lecture covered several supervised learning techniques for classification, methods for evaluating their performance, and the trade-offs involved.

- **Confusion Matrix** is used to quantify a classifier's performance using metrics like Sensitivity, Specificity, and Overall Accuracy
- **Receiver Operating Characteristic (ROC)** curve visualizes the trade-off between the True Positive Rate (Sensitivity) and the False Positive Rate (1 - Specificity) across all possible thresholds
- **Area Under the Curve (AUC)** provides a single aggregate score for model performance, where an AUC of 1 is a perfect classifier and 0.5 is equivalent to random guessing

- **Generative Models:** estimate the probability distribution of the predictors for each class and use Bayes' theorem.
  - **Linear Discriminant Analysis (LDA):** Assumes predictors follow a Gaussian distribution with a different mean for each class but a common covariance matrix. This results in linear decision boundaries
  - **Quadratic Discriminant Analysis (QDA):** More flexible than LDA, it assumes predictors follow a Gaussian distribution where each class has its own unique mean and covariance matrix. This results in quadratic (curved) decision boundaries

- **k-Nearest Neighbors (k-NN):** This is a non-parametric, instance-based algorithm that classifies a new data point based on a "majority vote" of its K closest neighbors in the training data
  - Choosing K: The choice of K involves a bias-variance trade-off. Cross-validation is used to find the optimal K

That's all Folks!